```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
import pandas as pd


xs=np.array([2,3,4,5,6,7,8,9])
ys=np.array([2.1,3.5,4.2,4.9,6.1,7.3,8.2,9.1])


#Modelo con NN

modelo= tf.keras.Sequential([tf.keras.layers.Dense(units=1,input_shape=[1])])
modelo.compile(optimizer="sgd",loss="mean_squared_error")



modelo.fit(xs,ys,epochs=10)
```

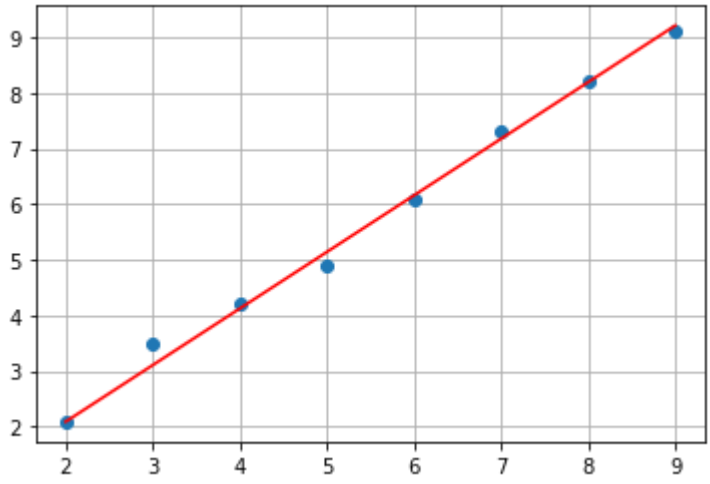
```
Epoch 1/10
1/1 [==============================] - 1s 800ms/step - loss: 3.4847
Epoch 2/10
1/1 [==============================] - 0s 11ms/step - loss: 0.2888
Epoch 3/10
1/1 [==============================] - 0s 13ms/step - loss: 0.0507
Epoch 4/10
1/1 [==============================] - 0s 12ms/step - loss: 0.0330
Epoch 5/10
1/1 [==============================] - 0s 17ms/step - loss: 0.0316
Epoch 6/10
1/1 [==============================] - 0s 14ms/step - loss: 0.0315
Epoch 7/10
1/1 [==============================] - 0s 15ms/step - loss: 0.0315
Epoch 8/10
1/1 [==============================] - 0s 12ms/step - loss: 0.0314
Epoch 9/10
1/1 [==============================] - 0s 13ms/step - loss: 0.0314
Epoch 10/10
1/1 [==============================] - 0s 28ms/step - loss: 0.0314
<keras.callbacks.History at 0x7fb70e417b50>
```

```
x_new=np.array([[10]])
y_pred=modelo.predict(x_new)
print(y_pred)
```

```
[[10.225656]]
```

```
y_nn=modelo.predict(xs)


plt.plot(xs,ys,"o")
plt.plot(xs,y_nn,"-r")
plt.grid()
plt.show()
```


9
8
7
6
5
4
3
2
2
3
4
5
6
7
8
9